Maybe, deep down, the universe has a less elegant structure than our experiences have led us to believe, or maybe we will find that our current aesthetic criteria need significant refining when applied in ever less familiar contexts. Nevertheless, especially as we enter an era in which our theories describe realms of the universe that are increasingly difficult to probe experimentally, physicists do rely on such an aesthetic to help them steer clear of blind alleys and dead-end roads that they might otherwise pursue. So far, this approach has provided a powerful and insightful guide.

In physics, as in art, symmetry is a key part of aesthetics. But unlike the case in art, symmetry in physics has a very concrete and precise meaning. In fact, by diligently following this precise notion of symmetry to its mathematical conclusion, physicists during the last few decades have found theories in which matter particles and messenger particles are far more closely intertwined than anyone previously thought possible. Such theories, which unite not only the forces of nature but also the material constituents, have the greatest possible symmetry and for this reason have been called supersymmetric. Superstring theory, as we shall see, is both the progenitor and the pinnacle example of a supersymmetric framework.

The Nature of Physical Law

Imagine a universe in which the laws of physics are as ephemeral as the tastes of fashion-changing from year to year, from week to week, or even from moment to moment. In such a world, assuming that the changes do not disrupt basic life processes, you would never experience a dull moment, to say the least. The simplest acts would be an adventure, since random variations would prevent you or anyone else from using past experience to predict anything about future outcomes.

Such a universe is a physicist’s nightmare. Physicists—and most everyone else as well—rely crucially upon the stability of the universe: The laws that are true today were true yesterday and will still be true tomorrow (even if we have not been clever enough to have figured them all out). After all, what meaning can we give to the term “law” if it can abruptly change? This does not mean that the universe is static; the universe certainly changes in innumerable ways from each moment to the next. Rather, it means that the laws governing such evolution are